Homework 9

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Problem 1. *Let*

$$\mathfrak{g} = \left\{ \begin{pmatrix} A & v \\ 0 & 0 \end{pmatrix} \mid A \in \mathfrak{gl}_3(\mathbb{R}), v \in \mathbb{R}^3 \right\} \subseteq \mathfrak{gl}_4(\mathbb{R}).$$

1. *Show that $\mathfrak{g}$ admits a decomposition*
$$\mathfrak{g} \cong \mathfrak{s} \ltimes \mathfrak{r},$$
where $\mathfrak{r}$ is the radical and $\mathfrak{s}$ is a semisimple Lie subalgebra. Determine exactly what $\mathfrak{s}$ and $\mathfrak{r}$ are.
2. *Let $K(x, y) = \operatorname{tr}(\operatorname{ad}_x \circ \operatorname{ad}_y)$ be the Killing form of $\mathfrak{g}$. Determine explicitly the Kernel*
$$\operatorname{Ker} K = \{x \in \mathfrak{g} \mid K(x, -) = 0\}.$$
3. *Using Cartan's criteria (assumed known):*
   (a) *Verify the criterion for semisimplicity on $\mathfrak{s}$.*
   (b) *Verify the corresponding criterion for semisimplicity on $\mathfrak{r}$.*
4. *Show that $K(\mathfrak{s}, \mathfrak{r}) = 0$.*

(1) We write $\mathbb{R}^3$ and $\mathfrak{gl}_3(\mathbb{R})$ to denote the relevant Lie subalgebras of $\mathfrak{g}$. The bracket on $\mathbb{R}^3$ is 0, and the bracket on $\mathfrak{gl}_3(\mathbb{R})$ is the commutator

$$\left[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \right] = \begin{pmatrix} [A, B] & 0 \\ 0 & 0 \end{pmatrix}.$$

We usually abuse notation, and write

$$A = \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 & v \\ 0 & 0 \end{pmatrix}, \qquad A + v = \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & v \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} A & v \\ 0 & 0 \end{pmatrix}.$$

Let $\mathfrak{z} \coloneqq \{c1 \mid c \in \mathbb{R}\} \subseteq \mathfrak{gl}_3(\mathbb{R})$, where 1 is the identity matrix. Let $\mathfrak{r}$ be the Lie subalgebra of $\mathfrak{g}$ whose underlying vector space is $\mathfrak{z} + \mathbb{R}^3$. Since $\mathfrak{z} \cap \mathbb{R}^3 \cong 0$, as vector spaces this is isomorphic to the direct sum $\mathfrak{z} \oplus \mathbb{R}^3$ (this does not promote to an isomorphism of Lie algebras), so we can write an element of $\mathfrak{r}$ as $c1 + v$, with $c \in \mathbb{R}$ and $v \in \mathbb{R}^3$. The bracket is simply the same as the bracket on $\mathfrak{g}$, meaning

$$[c1 + v, -] = [c1, -] + [v, -] = [v, -],$$

where we have used the fact that $c1$ is a diagonal matrix (viewed as an element of $\mathfrak{g}$, not $\mathfrak{gl}_3(\mathbb{R})$), and thus commutes with everything, so its commutator vanishes.

First, we show that $\mathfrak{r}$ is an ideal. Note that as vector spaces, we have $\mathfrak{g} \cong \mathfrak{gl}_3(\mathbb{R}) \oplus \mathbb{R}^3$, and since $\mathfrak{gl}_3(\mathbb{R}) \cap \mathbb{R}^3 \cong 0$, we have $\mathfrak{g} \cong \mathfrak{gl}_3(\mathbb{R}) + \mathbb{R}^3$. Thus we can similarly write an arbitrary element of $\mathfrak{gl}_3(\mathbb{R})$ as $A + v$. Now we compute

$$[A + v, c1 + w] = [A, w] + [v, w].$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & v_1 \\ 0 & 0 & 0 & v_2 \\ 0 & 0 & 0 & v_3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & a_{11}v_1 + a_{12}v_2 + a_{13}v_3 \\ 0 & 0 & 0 & a_{21}v_1 + a_{22}v_2 + a_{23}v_3 \\ 0 & 0 & 0 & a_{31}v_1 + a_{32}v_2 + a_{33}v_3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{R}^3 \subseteq \mathfrak{r}.$$

$$\begin{pmatrix} 0 & 0 & 0 & w_1 \\ 0 & 0 & 0 & w_2 \\ 0 & 0 & 0 & w_3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & v_1 \\ 0 & 0 & 0 & v_2 \\ 0 & 0 & 0 & v_3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = 0 \in \mathfrak{r}.$$

Therefore, $[\mathfrak{g}, \mathfrak{r}] \subseteq \mathfrak{r}$, so we have an ideal.

Now we show $\mathfrak{r}$ is solvable. We have shown that $[v, w] = 0$ for $v, w \in \mathbb{R}^3$, so

$$[c1 + v, d1 + w] = [v, d1 + w] = [v, w] = 0.$$

Thus $[\mathfrak{r}, \mathfrak{r}] \cong 0$ is solvable.

Now we show $\mathfrak{r}$ is the maximal solvable ideal of $\mathfrak{g}$. Since there is a maximal solvable ideal, let $\mathfrak{R} \subseteq \mathfrak{g}$ be the maximal solvable ideal, meaning $\mathfrak{r} \subseteq \mathfrak{R}$. We show that $\mathfrak{r} = \mathfrak{R}$.

Then since $\mathbb{R}^3 \subset \mathfrak{r} \subseteq \mathfrak{R}$, we must have that (as a vector space), $\mathfrak{R} = \left\langle \mathfrak{h} + \mathbb{R}^3 \right\rangle \cong \langle \mathfrak{h} \rangle + \mathbb{R}^3 \cong \langle \mathfrak{h} \rangle \oplus \mathbb{R}^3$ for some subset $\mathfrak{h} \subseteq \mathfrak{g}$. We can take $\mathfrak{h}$ to be a subspace of $\mathfrak{gl}_3(\mathbb{R})$ without loss of generality, such that $\mathfrak{h} = \langle \mathfrak{h} \rangle$. We see immediately that $\mathfrak{h} \supseteq \mathfrak{z}$.

We will use the fact that when not viewing $\mathfrak{gl}_3(\mathbb{R})$ as a subspace of $\mathfrak{g}$, and rather as the standalone Lie algebra, the solvable radical is $\mathfrak{z}$. We also use the fact that the embedding $\mathfrak{gl}_3(\mathbb{R}) \hookrightarrow \mathfrak{g}$ is a Lie algebra homomorphism, as noted at the beginning. Let $A + v \in \mathfrak{R}$ and $B + w \in \mathfrak{g}$. Then

$$[A + v, B + w] = [A, B]_{\mathfrak{gl}_3(\mathbb{R})} + [A, w] + [v, B] + [v, w].$$

As noted earlier, $[A, w], [v, B] \in \mathbb{R}^3$, and $[v, w] = 0$. Moreover, since $[A, B]$ is the Lie bracket in $\mathfrak{gl}_3(\mathbb{R})$, and since $\mathfrak{gl}_3(\mathbb{R})$ is a Lie algebra, we have $[A, B] \in \mathfrak{gl}_3(\mathbb{R})$. Therefore, $[A, B]$ is linearly independent from $[A, w] + [v, B]$, so this Lie bracket lands back in $\mathfrak{R}$ if and only if $[A, B] \in \mathfrak{R}$. We thus find that $\mathfrak{h}$ must be an ideal of $\mathfrak{gl}_3(\mathbb{R})$.

We now investigate the solvability of $\mathfrak{R}$. We have seen that the Lie bracket $[A + v, B + w]$ yields two linearly independent components: and $\mathbb{R}^3$-component and a $\mathfrak{h}$-component. Write $\mathfrak{R}_n$ for the nth term in the derived series.

We immediately have that

$$[\mathfrak{R}, \mathfrak{R}] = [\mathfrak{h}, \mathfrak{h}] + [\mathfrak{h}, \mathbb{R}^3] + [\mathbb{R}^3, \mathfrak{h}] + [\mathbb{R}^3, \mathbb{R}^3].$$

Induction then shows that

$$[\mathfrak{R}_n, \mathfrak{R}_n] = \mathfrak{h}_{n+1} + \mathbb{R}^3_{n+1} + \text{various bracktets between } \mathfrak{h} \text{ and } \mathbb{R}^3.$$

We denote the "various brackets between $\mathfrak{h}$ and $\mathbb{R}^3$" by $\mathcal{O}(n)$. Note that $\mathcal{O}(n) \subseteq \mathbb{R}^3$ always, since $[\mathfrak{h}, \mathbb{R}^3] \subseteq \mathbb{R}^3$. We now use linear independence of elements of $\mathbb{R}^3$ and elements of $\mathfrak{h}$. There must be N such that $\mathfrak{R}_N = 0$. By linear independence, $\mathfrak{h}_N = 0$ (beause you can't cancel the elements with elements in $\mathbb{R}^3 \supseteq \mathbb{R}^3_N + \mathcal{O}(N)$). In particular, $\mathfrak{h}$ must be solvable. Thus we have found that $\mathfrak{h}$ is a solvable ideal of $\mathfrak{gl}_3(\mathbb{R})$ such that $\mathfrak{z} \subseteq \mathfrak{h}$. However, $\mathfrak{z}$ is the maximal solvable ideal of $\mathfrak{gl}_3(\mathbb{R})$, so we conclude $\mathfrak{h} = \mathfrak{z}$, and thus $\mathfrak{R} = \mathfrak{r}$.

The obvious definition for $\mathfrak{s}$ is now $\mathfrak{s} := \mathfrak{sl}_3(\mathbb{R})$. We have seen that a solvable ideal of $\mathfrak{gl}_3(\mathbb{R})$ as a subset of $\mathfrak{g}$ must also be a solvable ideal of $\mathfrak{gl}_3(\mathbb{R})$ as a standalone Lie algebra, and the converse also obviously holds. Thus since $\mathfrak{sl}_3(\mathbb{R})$ is a semisimple Lie subalgebra of $\mathfrak{gl}_3(\mathbb{R})$, it must also be a semisimple Lie subalgebra of $\mathfrak{g}$.

We now show that as a vector space, $\mathfrak{g} \cong \mathfrak{s} \oplus \mathfrak{r}$. As vector spaces.

$$\mathfrak{s} \oplus \mathfrak{r} = \mathfrak{sl}_3(\mathbb{R}) \oplus (\mathfrak{z} \oplus \mathbb{R}^3) \cong (\mathfrak{sl}_3(\mathbb{R}) \oplus \mathfrak{z}) \oplus \mathbb{R}^3 \cong \mathfrak{gl}_3(\mathbb{R}) \oplus \mathbb{R}^3 \cong \mathfrak{g}.$$

Here we have used the Levi decomposition $\mathfrak{gl}_3(\mathbb{R}) \cong \mathfrak{sl}_3(\mathbb{R}) \ltimes \mathfrak{z}$. Since $\mathfrak{s} \cap \mathfrak{r} \cong 0$, and since $\mathfrak{g} \cong \mathfrak{s} + \mathfrak{r}$ as vector spaces, we have $\mathfrak{g} \cong \mathfrak{s} \ltimes \mathfrak{r}$ as Lie algebras.

(2) We need to find all $x \in \mathfrak{g}$ such that $\operatorname{tr}(\operatorname{ad}_x, \operatorname{ad}_y) = 0$. We first find $\operatorname{Ker}(x \mapsto \operatorname{ad}_x)$. We claim that $\operatorname{Ker} K = \mathbb{R}^3$. Note here that we have chosen to write K for the Killing form.

Let $x, y, z \in \mathfrak{g}$, and write

$$x = \begin{pmatrix} A & v \\ 0 & 0 \end{pmatrix}, \qquad y = \begin{pmatrix} B & w \\ 0 & 0 \end{pmatrix}, \qquad z = \begin{pmatrix} C & u \\ 0 & 0 \end{pmatrix}.$$

Then

$$\operatorname{ad}_y(z) = \begin{pmatrix} BC & Bu \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} CB & Cw \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} [B, C] & Bu - Cw \\ 0 & 0 \end{pmatrix}.$$

Then

$$\operatorname{ad}_x(\operatorname{ad}_y(z)) = \begin{pmatrix} [A, [B, C]] & ABu - ACw - [B, C]\, v \\ 0 & 0 \end{pmatrix}.$$

We can thus write ad_x as a $\mathfrak{gl}_3(\mathbb{R})$ component $\operatorname{ad}_x^{\mathfrak{gl}} : \mathfrak{gl}_3(\mathbb{R}) \to \mathfrak{gl}_3(\mathbb{R})$ given by $[A, -]$, a $\mathbb{R}^3$ component $\operatorname{ad}_x^{\mathbb{R}^3} : \mathbb{R}^3 \to \mathbb{R}^3$ which maps $w \mapsto Aw$, and an off-diagonal component $\operatorname{ad}_x^d : \mathfrak{gl}_3(\mathbb{R}) \to \mathbb{R}^3$ which maps $B \mapsto Bv$. We can write

$$\operatorname{ad}_x(B + w) = \begin{pmatrix} \operatorname{ad}_x^{\mathfrak{gl}}(B) & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \operatorname{ad}_x^{\mathbb{R}^3}(w) \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \operatorname{ad}_x^d(B) \\ 0 & 0 \end{pmatrix}.$$

If we give $\mathfrak{g}$ a basis $\{e_i\}_{1 \le i \le 12}$, then the trace of an operator $\mathcal{O} : \mathfrak{gl}_3(\mathbb{R}) \to \mathfrak{gl}_3(\mathbb{R})$ is given by

$$\operatorname{tr}\mathcal{O} = \sum_{i=1}^{12} \langle e_i, \mathcal{O} e_i \rangle,$$

where $\langle -, - \rangle$ is the dot product. In particular, this is nonzero if and only if $\mathcal{O} e_i$ can be written as a linear combination with a nonzero e_i component. Since ad_x^d maps $\mathfrak{gl}_3(\mathbb{R})$ into $\mathbb{R}^3$, and since these spaces have $\mathfrak{gl}_3(\mathbb{R}) \cap \mathbb{R}^3 \cong 0$, we see that ad_x^d is traceless, and thus does not contribute to the trace of ad_x. Here we note also that trace preserves addition.

Note that the map $\operatorname{ad}_x^{\mathbb{R}^3}$ mapping $w \mapsto Aw$ is just the linear map A, and thus the trace $\operatorname{tr}\left(\operatorname{ad}_x^{\mathbb{R}^3}\right) = \operatorname{tr} A$. Thus

$$\operatorname{tr}(\operatorname{ad}_{\mathrm{x}} \circ \operatorname{ad}_y) = \operatorname{tr}\left(\operatorname{ad}_x^{\mathfrak{gl}} \circ \operatorname{ad}_y^{\mathfrak{gl}}\right) + \operatorname{tr}(AB).$$

Since the trace thus depends entirely on the $\mathfrak{gl}_3(\mathbb{R})$ components of x, y, we see that $\mathbb{R}^3 \subseteq \operatorname{Ker} K$. Therefore, we adopt the convention of writing $\operatorname{ad}_x^{\mathfrak{gl}} = \operatorname{ad}_A^{\mathfrak{gl}}$ to make notation more clear.

We are looking for $x = A + v$ such that $K(x, -) = 0$; that is, for all $y = B + w \in \mathfrak{g}$,

$$\operatorname{tr}\left(\operatorname{ad}_A^{\mathfrak{gl}} \circ \operatorname{ad}_B^{\mathfrak{gl}}\right) + \operatorname{tr}(AB) = 0.$$

The first term is simply the Killing form $K^{\mathfrak{gl}}$ on $\mathfrak{gl}_3(\mathbb{R})$.